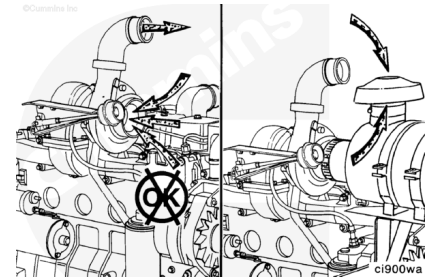


Trainings ▾

General Information

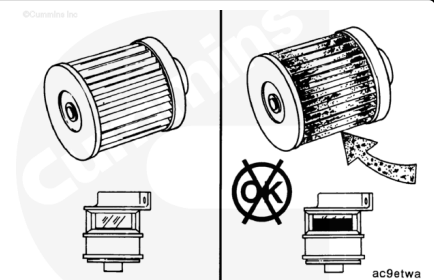
Diagnosing Air System Malfunctions

The correct amount of clean air to the cylinders is required for good performance. As discussed earlier in Airflow System, ingested dust and dirt will damage the engine cylinders. Dust and dirt can also damage the valve stems and guides, as well as wear down the turbocharger compressor vanes affecting efficiency. Larger debris can damage the blades of the turbocharger.



Airflow restriction results in excessive smoke and low power.

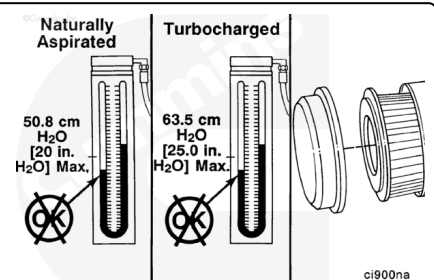
Restriction increases as the filter removes contaminants from the intake air. Restriction airflow changes the air-to-fuel ratio, reducing power and increasing smoke from the engine. Verify that the air cleaner is being maintained correctly.



Intake Air Restriction - Checking

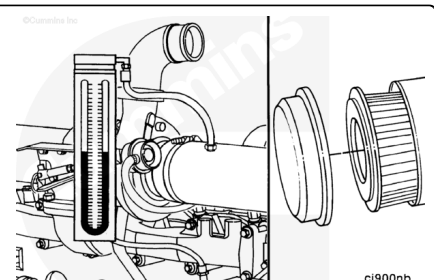
Replace the air cleaner element when the restriction reaches the maximum limit at rated engine power and speed.

Naturally Aspirated	Turbocharged
50.8 mm H ₂ O	63.5 mm H ₂ O
[2 in. H ₂ O]	[2.5 in. H ₂ O]



Note : The illustration shows the measurement in centimeters instead of millimeters.

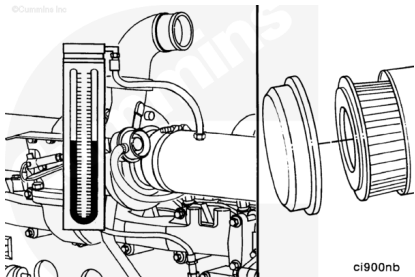
For turbocharged engines, measure the restriction just before the turbocharger. Measure just before the intake manifold for naturally aspirated engines.



Measure

Install a vacuum gauge or water manometer, Part Number ST-1111-3, in the intake air piping.

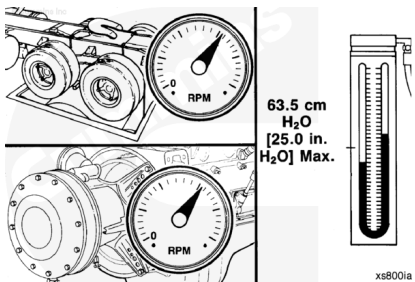
The gauge adapter **must** be installed at a 90-degree angle to the airflow in a straight section of pipe, one pipe diameter before the turbocharger.



Operate the engine at full throttle and rated rpm with maximum load.

Record the data on the gauge or manometer.

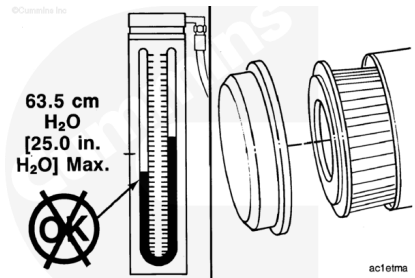
	mm-h2o		in-h2o	
Turbocharged		63.5	MAX	2.5
Naturally Aspirated		50.8	MAX	2



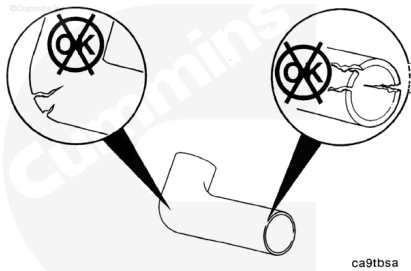
Note : The illustration shows the measurement in centimeters instead of millimeters.

If restriction exceeds specifications:

Replace or clean the air filter element. Refer to the equipment manufacturer's instructions.



Inspect the intake piping for damage. Refer to the equipment manufacturer's repair instructions.



Remove the test equipment.

